

```
# Step 1: Install gdown if needed
!pip install gdown --quiet

# Step 2: Download the file from Google Drive
import gdown

file_id = '1JfoBfxKwU11oshcp7Hk8ee987n3363ZP'
gdown.download(f'https://drive.google.com/uc?id={file_id}', 'complaints_sample.csv', quiet=False)

# Step 3: Load it using pandas
import pandas as pd

df = pd.read_csv('complaints_sample.csv')

# Step 4: Display first 5 rows to confirm it works
df.head()
```

Downloading...

From (original): <https://drive.google.com/uc?id=1JfoBfxKwU11oshcp7Hk8ee987n3363ZP>
From (redirected): [https://drive.google.com/uc?id=1JfoBfxKwU11oshcp7Hk8ee987n3363ZP&confirm=t&uuid=6ce37ee1-d7b9-400a-94ce-3510981d100%|5.59G/5.59G \[01:16<00:00, 73.1MB/s\]](https://drive.google.com/uc?id=1JfoBfxKwU11oshcp7Hk8ee987n3363ZP&confirm=t&uuid=6ce37ee1-d7b9-400a-94ce-3510981d100%|5.59G/5.59G [01:16<00:00, 73.1MB/s])
To: /content/complaints_sample.csv
<ipython-input-2-a28586f9391a>:13: DtypeWarning: Columns (16) have mixed types. Specify dtype option on import or set low_memory=False
df = pd.read_csv('complaints_sample.csv')

	Date received	Product	Sub-product	Issue	Sub-issue	Consumer complaint narrative	Company public response	Company	State	ZIP code	Tags	Consumer consent provided?	Submitted via
0	2025-04-09	Credit reporting or other personal consumer re...	Credit reporting	Improper use of your report	Credit inquiries on your report that you don't...	NaN	NaN	Experian Information Solutions Inc.	TX	78626	NaN	NaN	We
1	2025-04-10	Money transfer, virtual currency, or money ser...	International money transfer	Money was not available when promised	NaN	NaN	NaN	TransferWise Ltd	CA	92173	NaN	NaN	We
2	2025-04-11	Credit reporting or other personal consumer re...	Credit reporting	Improper use of your report	Reporting company used your report improperly	I have not supplied proof under the doctrine o...	Company has responded to the consumer and the ...	USCB Corporation	NC	28208	NaN	Consent provided	We
3	2025-04-11	Credit reporting or other personal consumer re...	Credit reporting	Problem with a company's investigation into an...	Investigation took more than 30 days	NaN	NaN	Experian Information Solutions Inc.	PA	19038	NaN	NaN	We
4	2024-06-11	Credit reporting or other personal consumer re...	Credit reporting	Improper use of your report	Reporting company used your report improperly	The Privacy Rule protects a consumer 's " nonp...	Company has responded to the consumer and the ...	Fidelity National Information Services, Inc. (...)	TX	77304	NaN	Consent provided	We

```
df.columns.tolist()
```

['Date received',
'Product',
'Sub-product',
'Issue',
'Sub-issue',
'Consumer complaint narrative',
'Company public response',
'Company',
'State',
'ZIP code',
'Tags',
'Consumer consent provided?',
'Submitted via',
'Date sent to company',
'Company response to consumer',
'Timely response?',
'Consumer disputed?',
'Complaint ID']

```
df.shape
```

```
(8796946, 18)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8796946 entries, 0 to 8796945
Data columns (total 18 columns):
#   Column                                Dtype
---  -
0   Date received                         object
1   Product                             object
2   Sub-product                         object
3   Issue                               object
4   Sub-issue                           object
5   Consumer complaint narrative         object
6   Company public response              object
7   Company                             object
8   State                               object
9   ZIP code                            object
10  Tags                                 object
11  Consumer consent provided?           object
12  Submitted via                        object
13  Date sent to company                 object
14  Company response to consumer         object
15  Timely response?                     object
16  Consumer disputed?                   object
17  Complaint ID                         int64
dtypes: int64(1), object(17)
memory usage: 1.2+ GB
```

```
df.isnull().sum().sort_values(ascending=False)
```

```
0
```

Tags	8189185
Consumer disputed?	8028642
Consumer complaint narrative	6010606
Company public response	4388913
Consumer consent provided?	1557639
Sub-issue	830255
Sub-product	235295
State	53203
ZIP code	30228
Company response to consumer	20
Issue	6
Date received	0
Product	0
Company	0
Date sent to company	0
Submitted via	0
Timely response?	0
Complaint ID	0

```
dtypes: int64(1)
```

```
df.sample(10)
```



	Date received	Product	Sub-product	Issue	Sub-issue	Consumer complaint narrative	Company public response	Company	State	ZIP code	Ta
3479135	2024-02-29	Debt collection	Credit card debt	Written notification about debt	Didn't receive notice of right to dispute	NaN	NaN	Resurgent Capital Services L.P.	VA	23664	Na
5784520	2021-08-19	Credit reporting, credit repair services, or o...	Credit reporting	Incorrect information on your report	Account status incorrect	NaN	Company has responded to the consumer and the ...	TRANSUNION INTERMEDIATE HOLDINGS, INC.	VA	20170	Na
5641467	2020-12-19	Credit reporting, credit repair services, or o...	Credit reporting	Incorrect information on your report	Information belongs to someone else	Dear XXXX XXXX XXXX XXXX, I request the credit...	Company has responded to the consumer and the ...	TRANSUNION INTERMEDIATE HOLDINGS, INC.	TX	XXXXXX	Older Americ
1276214	2025-01-15	Credit reporting or other personal consumer re...	Credit reporting	Incorrect information on your report	Public record information inaccurate	NaN	NaN	EQUIFAX, INC.	MI	48504	Na
8742293	2019-05-19	Debt collection	Other debt	Communication tactics	Frequent or repeated calls	NaN	Company believes it acted appropriately as aut...	Penn Credit Corporation	MD	21207	Na
3332378	2024-08-04	Credit reporting or other personal consumer re...	Credit reporting	Problem with a company's investigation into an...	Their investigation did not fix an error on yo...	NaN	NaN	General Motors Financial Company, Inc.	MI	480XX	Na
1633413	2025-02-06	Credit reporting or other personal consumer re...	Credit reporting	Problem with a company's investigation into an...	Their investigation did not fix an error on yo...	NaN	Company has responded to the consumer and the ...	Experian Information Solutions Inc.	CA	93722	Na
3396335	2024-11-05	Credit reporting or other personal consumer re...	Credit reporting	Incorrect information on your report	Account information incorrect	NaN	NaN	EQUIFAX, INC.	NY	10805	Na
7058825	2018-07-17	Credit reporting, credit repair services, or o...	Credit reporting	Incorrect information on your report	Information belongs to someone else	The following account needs to be deleted and ...	Company has responded to the consumer and the ...	Experian Information Solutions Inc.	GA	30350	Na
3834816	2023-07-16	Credit card or prepaid card	General-purpose credit card or charge card	Getting a credit card	Application denied	Citi denied a card after saying i i was pre-ap...	Company has responded to the consumer and the ...	CITIBANK, N.A.	FL	32205	Servicement

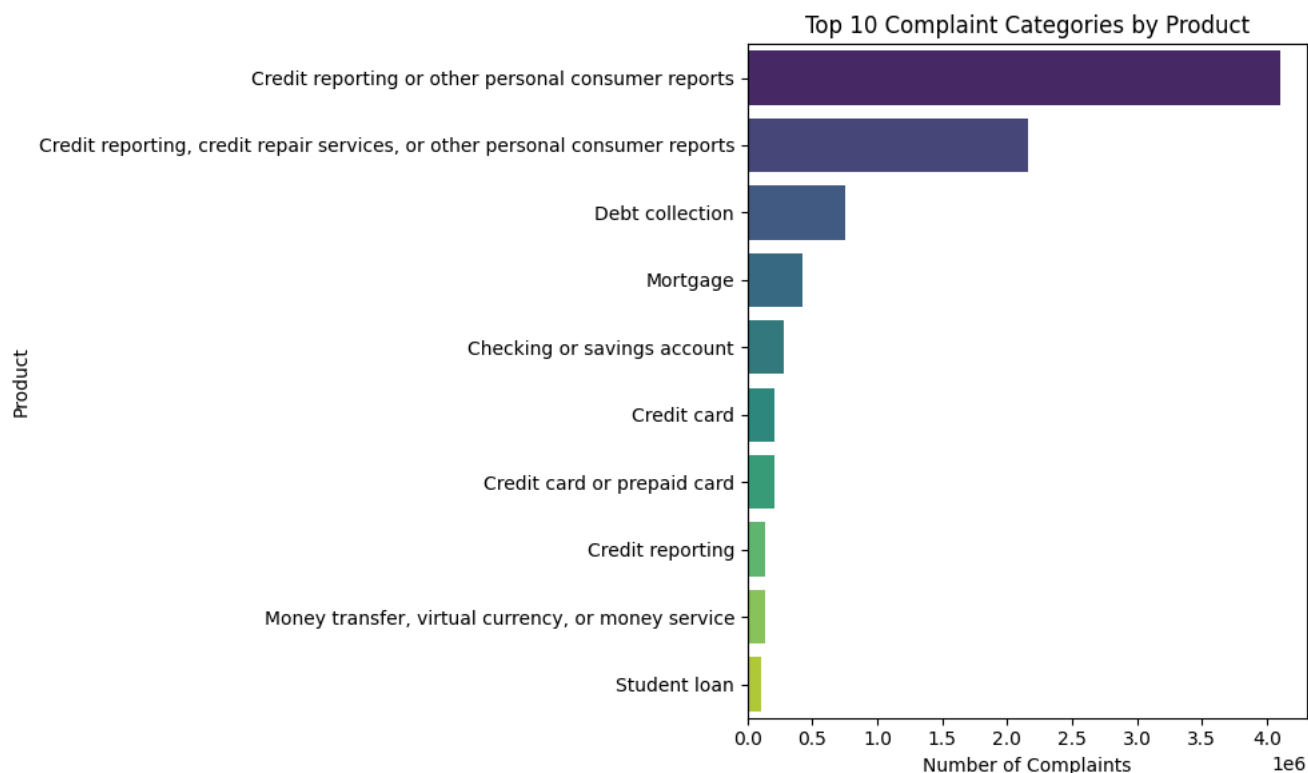
```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Top 10 most common product complaints
top_products = df['Product'].value_counts().nlargest(10)
```

```
plt.figure(figsize=(10,6))
sns.barplot(x=top_products.values, y=top_products.index, palette='viridis')
plt.title('Top 10 Complaint Categories by Product')
plt.xlabel('Number of Complaints')
plt.ylabel('Product')
plt.tight_layout()
plt.show()
```

 <ipython-input-8-068d0e1cce23>:8: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `l`
 sns.barplot(x=top_products.values, y=top_products.index, palette='viridis')



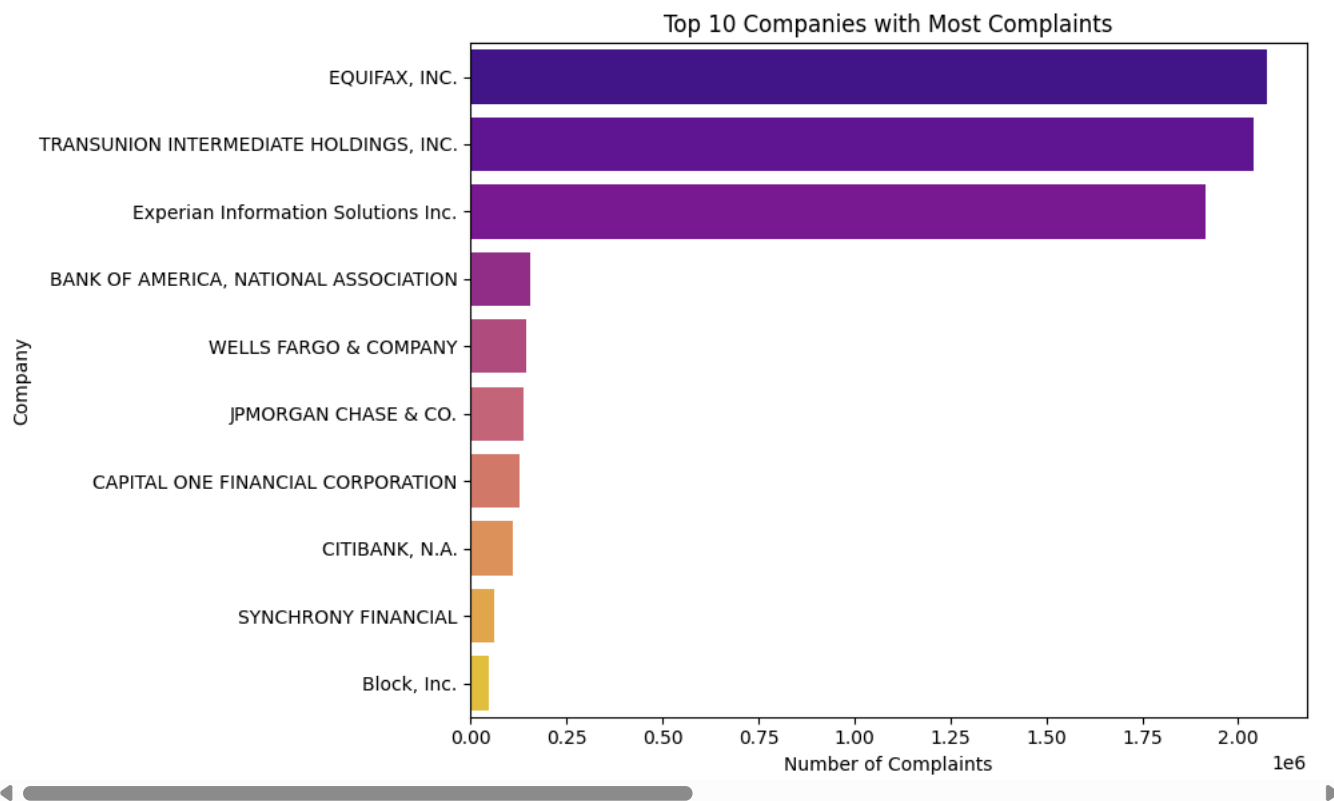
```
top_companies = df['Company'].value_counts().nlargest(10)
```

```
plt.figure(figsize=(10,6))
sns.barplot(x=top_companies.values, y=top_companies.index, palette='plasma')
plt.title('Top 10 Companies with Most Complaints')
plt.xlabel('Number of Complaints')
plt.ylabel('Company')
plt.tight_layout()
plt.show()
```

 <ipython-input-9-ee42a42fffb8>:4: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `le

```
sns.barplot(x=top_companies.values, y=top_companies.index, palette='plasma')
```



```
# Convert 'Date received' to datetime
df['Date received'] = pd.to_datetime(df['Date received'])
```

```
# Group by month
monthly_complaints = df.groupby(df['Date received'].dt.to_period('M')).size()
```

```
monthly_complaints.plot(kind='line', figsize=(12,6), title='Yearly Complaints Trend', color='teal')
plt.xlabel('Year')
plt.ylabel('Number of Complaints')
plt.grid(True)
plt.tight_layout()
plt.show()
```